

we talked to a very broad base of customers in India, the US, Japan and Europe, to identify the product segments where they wanted to operate. During those six months, we built a team rapidly, and made one acquisition to bring in some resources and skills,” says Kumar.

Even deep pockets do not guarantee success. Ultimately, semiconductor businesses must win customers, scale volumes and compete globally on cost and performance. Here, conglomerates retain an advantage—long-standing relationships with global firms open doors for initial engagements.

L&T Semiconductors is building the business with a high level of accountability where execution and outcomes are critical. “L&T is used to building very large systems, whether it is a ship or a missile or a tank, that sell for millions of dollars. Semiconductor is the opposite end of large in terms of how much you sell, as for a single chip, \$1 is a good price,” says Kumar.

CHOOSING THE RIGHT BATTLES

Today, NVIDIA is the global benchmark in semiconductor design, showing how ownership of chip architecture, software and IP creates outsize value. Indian conglomerates are not trying to replicate NVIDIA’s model nor are they attempting to compete with memory leaders such as Samsung Electronics or Micron Technology in advanced technologies such as high-bandwidth memory.

Instead, their initial focus is on semiconductor segments that are less complex, aligned with existing capabilities and offer clearer paths to customers and revenue. L&T, for instance, is focusing on these product areas: analog mix signals, radio frequency (RF), computing and power delivery.

Axiro Semiconductor is concentrating on high-performance RF, wireless infrastructure and satellite communication semiconductors. “These applications sit at the backbone of global connectivity, from 5G/6G radio units and fixed wireless access systems to low-earth orbit satellite terminals,” says Yanduru.

A similar approach is visible in the Tata Group. While Tata Consultancy Services (TCS) has collaborated with Tata Electronics on domestically designed chips, industry sources say these are primarily intended for internal applications, including at Tata Motors. Tata Electronics declined to comment and TCS did not respond to queries.

Starting with chips that are less complex increases



“Product is where the IP is, product is where the differentiation is. Once we sell enough products, we will build factories to make those products”

SANDEEP KUMAR
CHIEF EXECUTIVE,
L&T SEMICONDUCTOR
TECHNOLOGIES LTD



“Indian companies are tagged as service firms. To move out of that mindset, companies must create an IP or design a chip of their own”

NAVIN BISHNOI
INDIA COUNTRY MANAGER
AND AVP, MARVELL INDIA

the probability of success. Companies are focusing on these because they can enable this market at the fastest rate and serve a large volume of customers, says Bishnoi of Marvell India.

Given the time consumed to design chips, a company also has to ensure that the idea/chip remains relevant even after two-four years, because that’s what it will take to get the chip out and ready for someone to use, he adds. Crucially, these conglomerates are targeting distinct segments as no single company can address the market for a full range of semiconductor components.

As the India Semiconductor Mission 2.0 takes shape with a sharper focus on chip design and IP creation, the real test will be this: Can India back conglomerates willing to commit patience, capital and execution discipline that semiconductor design demands? **BT**